

ITIS4260 Project 2

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1 Project 2: Access Log Analysis

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[2]: # import pandas
import pandas as pd
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[4]: # read the datalog csv file into a dataframe and display a few rows
log_info_df = pd.read_csv('datalogUpdated.csv')
log_info_df.head()
```

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[4]:
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	Datetime	EmployeeID	Action	ResourceID	Access	IPAddress	Latitude	\
0	6/1/2023 0:05	10159	login	AuthServer		192.168.1.25	35.175927	
1	6/1/2023 0:09	10163	login	AuthServer		192.168.1.49	35.228719	
2	6/1/2023 0:10	10162	login	AuthServer		192.168.1.39	35.228719	
3	6/1/2023 1:14	10162	logout	AuthServer		192.168.1.39	35.228719	
4	6/1/2023 1:25	10163	logout	AuthServer		192.168.1.49	35.228719	

	Longitude	Result
0	-80.999608	success
1	-80.821720	success
2	-80.821720	success
3	-80.821720	success
4	-80.821720	success

```
[6]: # read the employee details csv file into a dataframe and display a few rows
employee_info_df = pd.read_csv('employee_details.csv')
employee_info_df.head()
```

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[6]:
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	EmployeeID	Name	Department	Start Date	End Date	Home	Latitude	\
0	10100	Adara	IT	5/21/14	12/30/20		35.191356	
1	10101	Addison	Finance	11/1/12	5/11/16		35.187293	
2	10102	Aimee	Research	11/16/14	12/30/20		35.216862	
3	10103	Althea	IT	3/11/14	12/30/20		35.261156	
4	10104	Amanda	Finance	5/24/13	12/30/20		35.255596	

	Home	Longitude
0		-80.842764

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1      -80.659253
2      -80.825005
3      -80.697252
4      -80.912228

```

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[8]: # read the department details csv file into a dataframe and display a few rows
dept_info_df = pd.read_csv('dept_resources.csv')
dept_info_df.head()

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[8]: ResourceID Department
0      Server0      HR
1      Server1      HR
2      Server2      HR
3      File0       HR
4      File1       HR

```

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[10]: # use the map function to match the employeeID with the employee name and add
      ↪ it to the datalog dataframe as a new column
log_info_df['Name'] = log_info_df['EmployeeID'].map(employee_info_df.
      ↪ set_index('EmployeeID')['Name'])

# use the map function to match the employeeID with the employee department and
      ↪ add it to the datalog dataframe as a new column
log_info_df['Employee Department'] = log_info_df['EmployeeID'].
      ↪ map(employee_info_df.set_index('EmployeeID')['Department'])

# use the map function to match the resourceID with the resource department and
      ↪ add it to the datalog dataframe as a new column
log_info_df['Resource Department'] = log_info_df['ResourceID'].map(dept_info_df.
      ↪ set_index('ResourceID')['Department'])

# display a few rows with the added columns
log_info_df.head()

```

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[10]: Datetime EmployeeID Action ResourceID Access IPAddress Latitude \
0 6/1/2023 0:05      10159  login AuthServer      192.168.1.25  35.175927
1 6/1/2023 0:09      10163  login AuthServer      192.168.1.49  35.228719
2 6/1/2023 0:10      10162  login AuthServer      192.168.1.39  35.228719
3 6/1/2023 1:14      10162  logout AuthServer      192.168.1.39  35.228719
4 6/1/2023 1:25      10163  logout AuthServer      192.168.1.49  35.228719

      Longitude Result      Name Employee Department Resource Department
0 -80.999608 success      Lenore      Finance      IT
1 -80.821720 success      Maisie      IT      IT
2 -80.821720 success MacKenzie      Finance      IT
3 -80.821720 success MacKenzie      Finance      IT
4 -80.821720 success      Maisie      IT      IT

```

```
[12]: # get the rows from the datalog dataframe where the employee department does not
      ↪match the resource department
result = log_info_df.loc[log_info_df['Employee Department'] !=
      ↪log_info_df['Resource Department']]

# display the employee name, department employee works for, resource accessed,
      ↪the department the accessed resource belongs to, date and time of access,
      ↪whether the access was successful or not
result = result[['Name', 'Employee Department', 'ResourceID', 'Resource
      ↪Department', 'Datetime', 'Result']]
result
```

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[12]:
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	Name	Employee Department	ResourceID	Resource Department	\
0	Lenore	Finance	AuthServer		IT
2	MacKenzie	Finance	AuthServer		IT
3	MacKenzie	Finance	AuthServer		IT
5	Lenore	Finance	AuthServer		IT
7	Amanda	Finance	AuthServer		IT
...	...	...	...	...	...
333944	MacKenzie	Finance	AuthServer		IT
333948	Stacy	HR	AuthServer		IT
333949	Denise	HR	AuthServer		IT
333950	Stacy	HR	AuthServer		IT
333951	Stacy	HR	AuthServer		IT

	Datetime	Result
0	6/1/2023 0:05	success
2	6/1/2023 0:10	success
3	6/1/2023 1:14	success
5	6/1/2023 1:56	success
7	6/1/2023 8:09	success
...	...	...
333944	6/29/2021 22:27	success
333948	6/29/2021 23:40	success
333949	6/29/2021 23:47	success
333950	6/29/2021 23:50	success
333951	6/30/2021 0:00	success

```
[96173 rows x 6 columns]
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